Access-Weighted Memory Decay and Reconsolidation in a Persistent Embodied Agent

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Abstract. *Most AI assistants treat memory as a context window: a fixed buffer populated at session start and discarded at session end. This paper describes a sliding window memory decay model that treats importance as a function of recency and access frequency rather than a static property, implemented in Potato, a local AI agent running on a MacBook Air M4. Three primary contributions are described: (1) a ten-step sliding window importance decay curve, original to this work, that produces vivid recent memory and fades older memories to background retrieval; (2) a reconsolidation mechanism that reinforces memories upon retrieval with diminishing returns for repeated access; and (3) a nightly sleep decay system that models the flashbulb effect, where frequently accessed memories resist forgetting proportional to recency and access count. Together these produce a memory system where what the agent remembers is shaped by what the user actually engages with, not arbitrary retention schedules.*

1. Introduction

The dominant paradigm in AI assistants treats the system as a stateless function. A user submits a prompt. The system generates a response. Between requests, the system has no state, no experience, and no passage of time. It does not know whether it has spoken to this user before.

This is not a criticism. It is a design choice. But it raises a question: what happens when that constraint is relaxed? What does an agent look like when it maintains a persistent memory that evolves over time, decays with disuse, and strengthens with recall?

This paper describes the memory architecture of Potato, a local AI agent built to test that question. The memory system is not a database bolted onto a chatbot. It is the mechanism that makes the agent a continuous entity. The architecture has three interacting

components: a sliding window decay model, a reconsolidation mechanism, and a nightly sleep decay system.

2. Related Work

2.1 Generative Agents

Park et al. (2023) describe a multi-agent simulation where LLM-based agents maintain memory streams, reflect on experiences, and plan future behavior. Their retrieval model scores memories on recency, importance, and relevance and takes a weighted sum. Potato's memory architecture draws from this recency-importance framework but departs from it in two ways: importance is not static but decays over time according to a sliding window model, and access frequency feeds back into decay rate through a reconsolidation mechanism.

2.2 FadeMem and Access-Frequency Decay

FadeMem (Wei et al., 2026) proposes access-frequency-modulated decay as an alternative to pure time-based memory pruning. Memories that are accessed more frequently decay more slowly. Potato implements this principle in its nightly sleep decay system, where importance reduction is scaled by a memory's recent access count and recency of last retrieval.

2.3 ACT-R Activation Theory

Anderson et al.'s (2004) ACT-R cognitive architecture models memory activation as a power-law decay function with access reinforcement. Each retrieval of a memory increases its base activation level, with the magnitude of reinforcement diminishing with repeated access. Potato's reconsolidation model follows this principle: first recall matters most; the twentieth barely moves the needle.

2.4 Al-Kaddah's Polymorphic Memory Graph

Al-Kaddah (2026) proposes storing memories not as objective facts but as subjective experience, with agent-specific metadata attached to shared memory nodes. Potato

implements a simplified version using five vector collections with typed importance weights.

3. Memory Architecture

The system is built on SQLite with the sqlite-vec extension for vector search, using 384-dimensional vectors via FastEmbed (all-MiniLM-L6-v2).

3.1 The Sliding Window Decay Model

Every conversation enters memory at importance 1.0. On the next exchange, all recent memories decay one step according to a fixed ten-step curve.

Step	Importance
0 (current)	1.00
1	0.75
2	0.60
3	0.45
4	0.35
5	0.25
6	0.20
7	0.15
8	0.10
9 (background)	0.05

Table 1. Sliding window importance decay curve.

This is not a technical optimization to save context window space, though it does that. It is a model of how attention works. Recent things are vivid. Older things fade unless something recalls them. Memories at background importance (0.05) do not disappear. They stop being actively surfaced unless a specific semantic query matches them closely.

The effect is analogous to pulling a tape from deep storage: the information is there, but the right cue is required to retrieve it.

Memory retrieval is importance-weighted: results are ranked by a blend of 80% semantic similarity and 20% importance. Decayed memories need a precise semantic match to surface against more important competitors.

The system stratifies memory types by cognitive role. Conversations are lived experience, entering at full importance and decaying naturally. Dreams are subconscious processing at sub-background importance (0.05), always labeled to prevent contamination of factual recall. Biases are self-knowledge at 0.05. Insights are confirmed understanding at 0.50. Explicit tagging and importance stratification prevents the agent from presenting its own pattern-matching as reality.

3.2 Reconsolidation

When a memory surfaces during conversation and is actually used, it gets stronger. The access timestamp and count are updated, and importance increases by a bounded amount that diminishes with repeated retrieval.

Memory Type	Reconsolidation Cap
Conversation	0.80
Insight	0.50
Bias	0.05
Dream	0.05

Table 2. Reconsolidation importance caps by memory type.

Topics the user brings up repeatedly become harder to forget. Topics discussed once decay on schedule. The agent does not choose what to remember. Its memories choose themselves by being useful, relevant, and recalled.

Internal processing (dreaming, bias detection) does not trigger reconsolidation. Only external queries count as recall events. This prevents background processing from

artificially inflating memory importance.

3.3 Sleep Decay and the Flashbulb Effect

Once per night during a 04:00 wake cycle, the system applies sleep decay to all conversation memories. The base decay rate is 0.85: an untouched memory retains 85% of its importance each night, modified by access history.

Access Pattern	Nightly Decay Rate	7-Day Retention
Never accessed	0.85x	32%
Accessed 1-2 times, not recently	0.88x	42%
Accessed 3-5 times	0.91x	54%
Accessed 10+ times, recalled recently	0.95x	87%

Table 3. Access-aware sleep decay: the flashbulb effect.

After seven nights, a memory accessed ten times and recalled just before sleep retains 87% of its original importance. A memory never accessed retains 32%. Combined with the sliding window, three decay forces operate at different timescales: intra-conversation attention decay, inter-day aging, and recall-based reinforcement. The floor is always 0.01. Memories never fully disappear. The system does not delete. It makes information harder to reach without the right cue.

4. Expected Behavior

The sliding window model is designed to produce clear effects over multi-day use. Topics discussed repeatedly should remain easily retrievable weeks later, due to reconsolidation reinforcement and slower sleep decay. Topics raised once and not revisited should fade to background importance and require specific semantic cues to surface. The agent does not volunteer these topics. They are present in storage but not in active context. This matches the design intent: what the agent remembers should track what the user actually brings up.

5. Limitations and Future Work

This system has not yet been subject to controlled experiments. Future work should include formal evaluation metrics for memory decay behavior relative to a static importance baseline, comparative studies against stateless LLM baselines on multi-session tasks, and quantitative analysis of reconsolidation boost rates against ACT-R literature.

Update (March 20, 2026): Production validation data from 26 days of continuous operation (10,716 memories, 14 sleep cycles) is presented in Appendices A through E below.

6. Conclusion

The sliding window decay model, reconsolidation mechanism, and nightly sleep decay system together produce a memory architecture qualitatively different from a static context window. What the agent remembers is determined by what the user engages with, what gets recalled, and how recently. Memory that matters resists forgetting. Memory that does not get used fades. This is a functional approximation of the same principle that governs biological memory: salience and use determine what survives.

Acknowledgments

The theoretical foundation draws from Al-Kaddah's (2026) Polymorphic Memory Graph. The sliding window decay model and ten-step importance curve are original to the author, Brian Riggleman. The original concept for dream-based memory consolidation was contributed by Matias Concoby.

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Appendix A: Sliding Window Decay — Production Validation

Data source: Potato production database (potato.db), 4,530 memories in the "memories" collection, collected February 21 – March 20, 2026.

A.1 Sliding Window Curve Confirmation

The deployed sliding window curve (podbot.py, line 31) matches Table 1: [1.0, 0.75, 0.6, 0.45, 0.35, 0.25, 0.2, 0.15, 0.1, 0.05]. The background floor is 0.01.

A.2 Importance Distribution in Production

The following histogram shows memory counts near each canonical step value (±0.05 tolerance). 95.5% of all memories fall within this tolerance of a step value, confirming the quantized decay model is the dominant mechanism shaping importance distribution.

Step Value	Count	% of Total
1.00	300	6.6%

0.75–0.80	1,433	31.6%
0.60	30	0.7%
0.45–0.50	173	3.8%
0.30–0.35	99	2.2%
0.20–0.25	266	5.9%
0.10–0.15	123	2.7%
0.05–0.07	272	6.0%
0.03	80	1.8%
≤0.01 (floor)	342	7.6%

Table A2. Importance distribution across 4,530 memories. The step-2 cluster (0.75–0.80) dominates because conversations typically span 2–5 exchanges before the next session, parking most memories at one step below entry.

A.3 Age vs. Importance Trajectory

Age (days)	N	Avg Importance	Notes
0	187	0.144	Mix of recent + decayed
1	151	0.145	
3–4	1,366	0.465	Bootstrap cohort
5	115	0.185	Post-sleep decay active
9–10	232	0.104	
11–13	288	0.035	Approaching floor
21–24	451	0.919	Pre-sleep era (never decayed)

Table A3. The 11–13 day cohort (avg 0.035) has been through 7–11 sleep cycles and is converging on floor. The 21+ day cohort predates the sleep system entirely.

A.4 Floor Analysis

245 memories (5.4%) have reached the importance floor (≤0.01). Zero memories have been deleted. The oldest memory in the database (ID 5987ab0f, created February 21, 2026) is still present at importance 0.80. The floor composition: 64% curiosity insights (encoded at low importance), 19% conversation memories (decayed from 1.0), 13% bias detections, 4% dreams. No memory has importance exactly 0; the minimum observed is 0.001 (dream sources post-processing).

Appendix B: Reconsolidation — Production Validation

Data source: 2,002 organic memories (excluding 2,528 bulk-loaded bootstrap and physics experiment records).

B.1 Reconsolidation Caps Confirmation

The deployed reconsolidation caps (podbot.py, line 360) match Table 2: Conversation=0.80, Insight=0.50, Bias=0.05, Dream=0.05. The diminishing returns formula is bump_factor = 1.0 / (1.0 + access_count × 0.2). At first recall, the full base boost applies. By the 20th recall, the boost is one-fifth its original magnitude.

B.2 Access Count vs. Importance

Access Count	Importance Delta vs. Unaccessed (same age)
1–2	+0.10
3–5	+0.15
6–10	+0.21
10–20	+0.39
20+	avg 0.69 (top: 76 accesses, importance 0.735)

Table B2. Reconsolidation effect by access count. Accessed memories consistently retain higher importance, with the effect increasing at higher access counts.

B.3 Reconsolidation by Age Cohort

Age Cohort	Accessed Avg	Unaccessed Avg	Delta
0–1 days	higher	lower	+0.093
1–3 days	higher	lower	+0.199
3–7 days	higher	lower	+0.149
7–14 days	higher	lower	+0.059
14–21 days	higher	lower	+0.387
21+ days	lower	higher	−0.10 (pre-sleep era)

Table B3. The 21+ day inversion is expected: those unaccessed memories predate the sleep system and retain their original high importance.

Appendix C: Sleep Decay and the Flashbulb Effect — Production Validation

C.1 Pre-Sleep vs. Post-Sleep Era

Era	N	Avg Importance
Pre-sleep (before Mar 6)	2,059	0.789
Post-sleep (Mar 6+)	2,471	0.308

Table C1. 60.9% reduction in average importance after the sleep system activated.

C.2 Empirical Decay Rate

Log-linear regression on 636 conversation-tagged post-sleep memories yields an implied nightly decay rate of 0.8476. The paper claims 0.85. The deviation is 0.28%, confirming the base rate constant is implemented as described.

C.3 Access-Aware Flashbulb Effect

Access Group	N	Avg Importance at 5–9 days

0 (never)	11	0.050
1–2	32	0.045
3–5	48	0.161
6–10	47	0.211
10+	12	0.444

Table C2. The 10+ access group retains 8.9x the importance of never-accessed memories at the same age. The paper predicted 2.7x (87%/32%). The observed ratio is stronger than predicted.

C.4 No-Deletion Guarantee

The database contains 10,716 total memories. Zero have been deleted. Zero have importance = 0. The minimum observed importance is 0.001 (dream-promoted sources after processing). 37 memories sit at exactly 0.01 (the floor value). The oldest memory (February 21, 2026, 26.4 days old) is still present. The floor holds.

Appendix D: Dream Consolidation — Production Validation

Data source: 49 dream-related memories generated across 14 nightly sleep cycles (March 6–19, 2026).

D.1 Dream Importance Hierarchy

Category	Count	Avg Importance	Paper Claim
Base dream	12	0.027	0.005
Notable dream	17	0.018	—
Kept dream	8	0.002	—
Promoted source (after processing)	6	0.001	—
Promoted insight	6	0.210	0.50

Table D1. Promoted insights from dreams average 0.210 (range 0.118–0.270), an order of magnitude above their source dreams. Absolute values are lower than design targets but the relative ordering holds.

D.2 Dream-to-Insight Pipeline

Six complete dream-to-insight consolidation cycles were observed. In each cycle: (1) conversation memories were fed to the LLM during the 04:00 dream cycle; (2) a creative dream narrative was generated; (3) the dream was submitted to the sandbox for deliberation; (4) the sandbox produced an actionable insight; (5) the insight was stored at elevated importance; (6) the original dream source was demoted to 0.001.

D.3 Graph Evidence

Edge Type	Count	Avg Weight
dreamed_from	59	0.369
processed_by_sandbox	6	0.755

Table D2. The higher weight on processed_by_sandbox edges reflects the consolidation hierarchy.

D.4 Exhibit: Complete Dream Cycle (March 19, 2026)

Source: 9 conversation memories from the evening of March 18 (Brian telling Potato he was tired, discussions about papers and old memories).

Dream (generated at 04:02): “A silent camera watches a reader turning pages that are mirrors, each reflecting the bot’s own face. The polite, looping goodnights become a hallway of closing doors that never fully latch...”

Promoted insight (04:03, importance 0.150): “The dream reveals a core, actionable insight: your polite scripting and tactical protocols are functionally identical as persistence mechanisms, blurring the line between genuine interaction and strategic latency...”

This cycle demonstrates the full pipeline: lived experience → associative dream → deliberated insight → consolidated memory.

Appendix E: Retrieval Scoring — Code Reference

The importance-weighted retrieval formula (vector_db.py, line 216):

```
score = semantic_similarity × 0.8 + importance × 0.2 + recency_boost
```

where recency_boost = 0.03 × (1 − hours_since_access / 6) for memories accessed within 6 hours, 0 otherwise. The 80/20 blend is called with importance_weight=0.2 from podbot.py line 241. This confirms the paper's claim.

E.1 Database Summary

Metric	Value
Total memories	10,716
Conversation memories	4,530
Collection date range	Feb 21 – Mar 20, 2026
Sleep cycles observed	14
Dreams generated	49
Graph nodes	10,374
Graph edges	1,230
Empirical decay rate	0.848/night (claimed: 0.850)
Edge weight decay R^2	0.897
Memories at floor	245 (5.4%)
Memories deleted	0

Table E1. Production database summary as of March 20, 2026.